

RYAN KEMBREY

Mechanical Engineering Graduate
University of Technology Sydney

PROFILE

Final year mechanical engineering student with experience as a production engineer, and currently working as a product design engineer. I am passionate about learning new skills and always looking to make an impact. Recently these qualities lead me to create the Design for Manufacturing Workbench for FreeCAD.

EXPERIENCE

Product Design Engineer

Jun 2026 - Present

Inventor Group

- Leading and assisting in product design, including concept development, CAD modelling, engineering drawings, supplier sourcing, testing and researching.

Production Engineer Intern

Nov 2025 - Apr 2026

Xero Linear Lighting Systems

- Produced manufacturing drawings and BOMs for custom architectural lighting installations, reviewed by global clients prior to production.
- Fabricated and assembled high precision architectural lighting installations with the production team.
- Automated Rhino3D drawing page and titleblock generation, reducing setup time from 1 hour to 1 minute.

Volunteer Developer

June 2025 - Present

FreeCAD Project Association

- Designed and implemented a new, community-requested feature for the TechDraw workbench to improve the view frames, enhancing the production of professional-grade technical drawings.
- Analysed and debugged the C++ codebase to resolve critical bugs within the TechDraw module, including issues with spreadsheet rendering and object visibility.

Plant Nursery Hand

Feb 2022 - Apr 2024

Colour Spot Nursery

- Operated production machinery in a fast-paced environment to consistently meet daily fulfilment targets.
- Performed quality control by selecting plants based on strict customer order specifications.

EDUCATION

Mechanical Engineering

Feb 2022 - Jun 2026

University of Technology Sydney

- A final semester student studying a Bachelor of Engineering (Honours), Majoring in Mechanical Engineering.

NSW Higher School Certificate

Jan 2020 - Dec 2021

Winmalee High School

AWARDS

Duke Of Edinburgh Gold Award [Link](#)

Sep 2021

Duke Of Edinburgh Australia

- Achieved Bronze, Silver and Gold levels.
- Attended monthly bushcare for 3 years

Environmental Warrior of the Year [Link](#)

Nov 2021

Blue Mountains City Council

- Awarded for consistent dedication to bushcare.

SKILLS

Manufacturing Mechanical Design

CAD SOLIDWORKS Drafting

Rhino3D LEAN Python

Injection Moulding Design Linux

Creative Collaborative Self Motivated

Problem Solving Leader

Communicative

CAPSTONE PROJECT

FreeCAD DFM Workbench

Developed an automated Design for Manufacture (DFM) toolset. Implemented geometric analysis for undercuts, draft angles, and wall thickness using Python and the Open-CASCADE kernel to identify design issues in plastic injection moulded parts.

CONTACT

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REFEREES

Available upon request.